

Although Google is the clear market leader in cloud computing services, it does have competitors. Identify one or two competitors in each cloud market segment—SaaS, PaaS, and IaaS—and compare their services with Google’s. In which market segment is Google most dominant? In which is it least dominant?

SaaS (Software as a Service)

The main SaaS that Google offers is Google Workspace, which consists of extremely popular and widely used software such as Gmail, Drive, Docs, and Meet. Google’s main rivals in terms of software services are Microsoft 365 and Salesforce. Since Microsoft 365 is more of a direct competitor to Google, we’ll solely focus on it as we compare the two financially. Google Workspace has great simultaneous and real-time collaboration integration. That is to say that collaborating with other people on services like Docs or Sheets is made seamless and efficient to set up. The reason for this is that Google takes a cloud-first approach to SaaS. Microsoft 365, on the other hand, prioritizes depth and control over the software services that it offers. A good example of this is Microsoft Excel compared to Google Sheets. Excel offers much more functionality, macros, and offline ability than Google Sheets. Excel (as well as other Windows programs) has deep Windows integration. This becomes especially useful when working in large enterprise environments that rely on the broader Microsoft 365 ecosystem. Although the same can be said for working in environments that rely on Google Workspace rather than Microsoft 365, the point still stands that Microsoft simply does not follow Google’s cloud-centered approach to SaaS. This is most clearly seen when looking at the history of Microsoft software. Microsoft 365 originally came from Microsoft Office, a suite of programs native to the Windows operating system that was dominant until the wide-spread adoption of cloud-era computing in the mid to late 2010s. Google’s Workspace is the more affordable of the two options which makes it the most popular among small businesses and teams. Microsoft 365 makes up for lack of affordability with it being the industry standard for enterprise software.

Google Workspace	\$6-18/person/month
Microsoft 365	\$30/person/month

PaaS(Platform as a Service)

Opposite to the dynamic of Google Workspace and Microsoft 365, Google has a competitive edge in terms of PaaS. Google's competitors in this field are both Microsoft and Amazon, with Azure and AWS (Amazon Web Services) respectively. The largest contributors to Google's domination in this case are Kubernetes and internal AI research.

Kubernetes

Kubernetes is an open-source containerization system that automates the deployment, scaling, and management of containerized applications. Containerized applications (and containerization in general) is the process of bundling applications alongside everything they need to run (code, dependencies, libraries, etc.) One of, if not the biggest advantages to containerized applications is that they essentially provide the same runtime safety* and consistency that virtual machines do at a fraction of the start time and resources. Google originally developed Borg as a proprietary solution for the same problem that Kubernetes would solve later: managing massive containerized applications for Google's massive data centers. Google kept Borg entirely internal for a long time which is what kept them on top of data center infrastructure, but with containerization going mainstream in 2013-2014 (largely due to Docker), they decided to release Kubernetes, which was built off of Borg.

**In terms of portability and consistency that is. Considering safety, VMs are much more secure.*

AI Research

Google has been conducting cutting-edge research, both academic and applied, on AI and machine learning since the early 2000s. The effects of this research are best seen with papers like "Attention is All You Need", a landmark paper conducted by researchers working on the Google Brain project, which was later succeeded by Google Deepmind and more generally Google AI. The paper introduced the transformer, a neural model used by virtually every natural language model today, including OpenAI's implementation of the GPT (generative pretrained transformer). In the context of PaaS specifically, Google has developed Vertex AI, a machine learning platform that utilizes Google's decades of AI research and allows users to

build, deploy, and manage AI models at scale. AI infrastructure is especially challenging for researchers, data scientists, and developers to maintain and scale affordability which makes Vertex AI that much more useful.

IaaS(Infrastructure as a Service)

Infrastructure as a service is where Google most clearly lacks in terms of market share and overall usage. AWS holds 29-30% of the market, followed by Azure with 20-22%, and in third place Google holds 12-13%. The reason for Amazon's clear control of the market in this case is that Amazon was the first mover in this field. They first launched their IaaS in 2006, providing cloud infrastructure on demand. This move gave them much more time to establish themselves in the market before Google or Microsoft could start to compete. When entire enterprises commit to one ecosystem of infrastructure, the cost of switching to another provider starts to compound as more services are employed. Because Amazon was the first to provide these services, there was a considerable interval of time where they were the only option, which meant that large companies were locked in economically to AWS. While Google has made significant steps in closing the market gap in IaaS (namely in data analysis and machine learning), the process remains difficult and slow.

Google holds a strong position in all three aforementioned segments of cloud computing, but the degree of dominance varies across those segments. In SaaS, Google faces its most competitive challenge from Microsoft 365. In PaaS, the competition is evenly balanced between Microsoft and Amazon, with the three tech giants constantly fighting for dominance. Lastly, for IaaS, we see that Google is at its weakest among its competitors, but they are putting in a noticeable effort to change that.